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Precise Point Positioning Performance Evaluation of QZSS Centimeter Level Augmentation Service

Maosen Hao¹, Wenhai Jiao², Xiaolin Jia³, and Qingrui Tao¹

Abstract. The Japanese Quasi-Zenith Satellite System(QZSS) provides Centimeter-Level Augmentation Service(CLAS) through L6 signal. The L6 augmentation messages includes satellite orbit correction, clock correction, phase bias, code bias, troposphere correction, ionosphere correction and other augmentation messages. These augmentation messages are used for a PPP-RTK service. Using PPP-RTK service, users in Japan can get high-precision real-time positioning results. Based on the measured data of AIRA, STK2, TSK2 and USUD of four MGEX stations in Japan, this paper analyzes the PPP augmentation performance of L6 augmentation signal by post static PPP, and compares it with the calculation results of QZSS final orbit and clock products. The results show that there is still a certain gap between the accuracy of orbit and clock augmented by L6 signal and the final products of QZSS. The static PPP based on L6 signal can realize centimeter-decimeter level positioning, and the positioning accuracy is lower than that of the final products of QZSS. Analysis reason: as CLAS are used to augment PPP-RTK services, PPP augmented service is out of scope of its pre-service. Therefore, the published orbit and clock correction are not a real-time precise orbit and clock correction, which leads to a gap between the corrected orbit and clock and the final products. Similarly, due to different application services, there are differences in positioning results.

Keywords: Centimeter-Level Augmentation Service • PPP • L6 signal • Signal-in-Space Accuracy • SSR

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1 Introduction

Quasi-Zenith Satellite System (QZSS) is a new generation of regional satellite navigation system jointly established by the Japanese government and enterprises. On November 1, 2018, the Japanese government announced the official operation of the QZSS system. The QZSS system consists of four satellites, three are Inclined Geo-Synchronous Orbit (IGSO) satellites, and one is a Geo-Synchronous Orbit (GEO) satellite. There are two types of augmentation services officially provided by QZSS: Sub-meter Level Augmentation Service (SLAS) and Centimeter-Level Augmentation Service (CLAS), the augmented services are realized by broadcasting L1-SAIF augmented information and L6 augmented information respectively [1]. The CLAS is mainly applied in Japan to achieve high-precision positioning services. The L6 signal is divided into L61 and L62 signals. The L61 signal is transformed from the LEX signal of the former QZSS-1 star (MICHIBIK-I), and the L62 is mainly used for the QZS 2-4 star. At present, L62 information is divided into two types: L6D and L6E. L6D is mainly used for the CLAS service, and L6E is mainly used for CLAS-E (Centimeter-Level Augmentation Message for Experiments) service. It is released by Japan JAXA (Aerospace Exploration Agency) Multi-GNSS Advanced Demonstration tool for Orbit and Clock Analysis (MADOCA) products based on L6E signals. The L6 signal herein refers to the L6D signal for CLAS service.

QZSS mainly releases augmented information in the form of RTCM-SSR, including correction information of satellite orbit and clock error, phase bias, code bias, ionosphere, etc. Users can receive augmented information and broadcast ephemeris in real-time, extract and correct augmented information. Corrected to the broadcast orbit to generate a more accurate real-time satellite orbit and clock offset to achieve high-precision PPP. During the development of the QZSS system, many experts and scholars studied the QZSS system from different aspects. Wang Guohui analyzed that the orbit error of the MICHIBIK-I satellite is better than 11cm, the clock error is better than 16.8cm [2]; Suelynn Choy et al. [3] proved that the performance of PPP based on LEX can reach decimeter level in Japan and Australia; Lou Yidong and others proved the feasibility of using the LEX signal to achieve sub-meter PPP positioning in the eastern coastal areas of China [4]; Zhang Wenfeng and Jiang Yongsheng found that kinematic and static PPP solutions based on LEX can achieve decimeter and centimeter accuracy, respectively [5-6]; Bu Weijin found a combination of BDS and QZSS positioning can improve the positioning accuracy, reliability, and stability of BDS single systems in the Asia-Pacific region [7]; Robert Odolinski et al. [8] found that at higher cut-off altitude angles, the four-system (BDS, Galileo, GPS, QZSS) RTK model allows for improved integer ambiguity resolution and positioning performance over the single-, dual- or triple systems; Hiromune Namie, etc. found that the Network RKT post-positioning accuracy based on LEX signals can reach RMS and STD within 1cm [9]; Zhang Shaocheng et al. [10] proved that real-time

L6E products can provide orbit and clock accuracy of centimeter level, and kinematic PPP positioning can reach 4.9cm, 4.2cm, 11.7cm in three directions of ENU, respectively; Po Chunhui et al. [11] found that L6D-based single-frequency PPP can improve the convergence time. Within a day, 83% of epoch positioning accuracy can reach 1m, and 50% of epoch positioning accuracy can reach 0.3m. This paper mainly introduces the CLAS augmented service based on QZSS L6D signals, and uses the measured data to evaluate the SIS User Range Error of the augmented orbit and clock offset. At the same time, the dual-frequency PPP performance of the augmented service is evaluated.

2 L6 augmentation information content and application

2.1 L6 augmentation information introduction

QZSS system transmits a total of 5 frequencies and 8 signals, including GPS-compatible PNT service signals: L1C/A, L1C, L2C and L5; L1S signal for the fully functional sub-meter level augmentation service (SLAS) and disaster and crisis management information dissemination service (DC Report); Provide L5S signal of satellite positioning new technology experiment service (PTV); L6 signal for centimeter-level augmentation service (CLAS). CLAS augments the service by broadcasting L6D augmentation information on L6 frequency band. The service area of CLAS is in Japan, which is divided into 19 networks. The service scope and Network distribution are shown in Fig. 1, different colors represent different Network areas [11]. The center frequency of the L6 signal is 1278.75 MHz and the frequency bandwidth is 39.0 MHz. The four QZSS satellites (J01, J02, J03, and J07) all provide centimeter-level augmentation service. L6 signals are stored in SSR format using RTCM 10403.2 standard. L6 files are published hourly and can be downloaded from the official QZSS website (<https://sys.qzss.go.jp/dod/en/archives/agree.html>).

Table 1. Correction information in L6 document

Message Sub Type ID	Message Name	Nominal Update Interval(s)
1	Compact SSR Mask	30
2	Compact SSR GNSS Orbit Correction	30
3	Compact SSR GNSS Clock Correction	5
4	Compact SSR GNSS Satellite Code Bias	30
5	Compact SSR GNSS Satellite Phase Bias	30

6	Compact SSR GNSS Satellite Code and Phase Bias	30
7	Compact SSR GNSS URA	30
8	Compact SSR STEC Correction	30
9	Compact SSR Gridded Correction	30
10	Compact SSR Service Information	(N/A)
11	Compact SSR GNSS Combined Correction	5 or 30

Table 1 shows the correction information contained in the L6 file. At present, correction information of GNSS satellites only includes correction information of GPS satellites, Galileo satellites and QZSS satellites. GLONASS system and BDS system are under construction. Type 6 is the correction information of satellite code bias and phase bias. The user selects the applicable correction information according to the Network ID of the area in which they are located. The ionospheric correction information and tropospheric correction information in types 8 and 9 are limited to use in Japan. The ionospheric grid information and tropospheric correction information suitable for this region should be matched with the Network ID in Japan. Type 11 combined correction information contains orbit correction information and clock correction information.

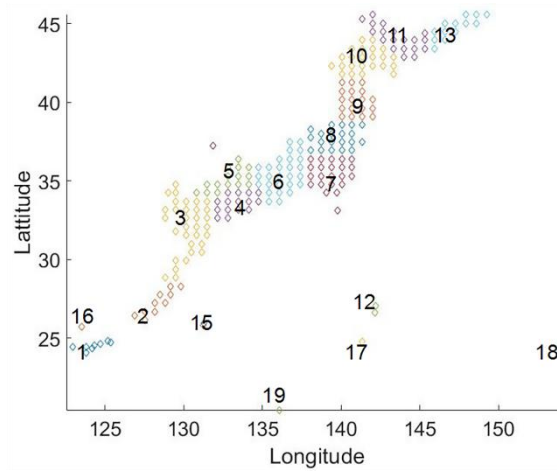


Fig. 1. CLAS service area and Network distribution

2.2 Real-time precise orbit and clock error recovery

The orbit and clock offset correction information in the L6D augmentation information is added to the orbit and clock offset calculated by the broadcast

ephemeris to obtain the recovered real-time precise orbit and clock offset. The orbit correction data in Table 1 are the correction information in the orbit coordinate system, which indicate the satellite position correction data in three directions: radial, along-track and cross-track. The correction information are based on the satellite antenna phase center (APC, Antenna Phase Center); clock error correction data is based on the correction information of clock error parameters in broadcast ephemeris: C_0 , C_1 , C_2 , where C_1 and C_2 is zero. The time reference of each correction information is GPS time system.

CLAS augmentation algorithm stipulates: if the Network ID of type 11 correction information in Table 1 is the same as the Network ID of the user's location, then the correction information of orbit and clock in type 11 correction information shall be adopted for correction. Otherwise, the correction information of orbit and clock in the correction information of type 2 and type 3 are used to correct.

Real-time precise orbit recovery:

Suppose the orbit correction information at time t is (R_r, R_a, R_c) , the satellite position in the ECEF coordinate system calculated by the broadcast ephemeris at the moment is $R = (X_0, Y_0, Z_0)$, and the satellite velocity is $V = (V_x, V_y, V_z)$, and the correction information in the ECEF coordinate system is $(\Delta X, \Delta Y, \Delta Z)$. Firstly, the modified value is transformed from orbital coordinate system to ECEF coordinate system:

$$\begin{pmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{pmatrix} = (e_r, e_a, e_c) \begin{pmatrix} R_r \\ R_a \\ R_c \end{pmatrix} \quad (2.1)$$

$$\text{Where: } e_a = \frac{V(t)}{|V(t)|}, e_c = \frac{R(t) \times V(t)}{|R(t) \times V(t)|}, e_r = e_a \times e_c.$$

Then the satellite position corrected at time t is:

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} X_0 \\ Y_0 \\ Z_0 \end{pmatrix} + \begin{pmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{pmatrix} \quad (2.2)$$

Real-time precise clock error recovery:

Suppose the clock difference correction parameter at time t is (C_0, C_1, C_2) , and the correction parameter C_1 and C_2 in QZSS are zero, the satellite clock difference calculated by broadcast ephemeris at this time is t_g , and the speed of light is c , then the corrected satellite clock difference is t_s :

$$t_s = t_g - \frac{C_0}{c} \quad (2.3)$$

3 Accuracy analysis of real-time orbit and clock error

In order to evaluate the GPS real-time orbit and clock offset data generated by the L6 augmented information, GPS precise orbits with an epoch interval of 15 min and precise clock offset with an epoch interval of 5 min provided by QZSS were selected as references (<https://sys.qzss.go.jp/dod/en/archives/pnt.html>). Relevant data of DOY 258-260 days in 2019 were selected for experiments to analyze the augmented orbit and clock difference accuracy.

Assessment methods:

The space-time datum is unified, and the time system is unified to GPST, and the coordinate datum is unified to ITRF2008 coordinate system.

Antenna phase center conversion: since the reference point of precise orbit satellite position is the satellite center of mass, the reference point of broadcast ephemeris and L6 correction information is the satellite antenna phase center. Therefore, when comparing the accuracy of the orbit, the IGS14 antenna file was used to correct the orbit coordinates from the antenna phase center to the satellite mass by using the dual-frequency ionospheric combination. Then compare the orbit R, A, and C to get the errors of the augmented orbit and QZSS precise orbit in the R, A, and C direction, and then calculate the RMS values in each direction.

Clock offset accuracy analysis: In order to eliminate the influence of different clock offset benchmarks on the accuracy analysis, this paper adopts the quadratic error method to analyze the corrected satellite clock offset. The specific methods are as follows: first, make a difference between the satellite clocks with the same epoch to obtain a difference. Then sum up the difference between the epochs of the same satellite and calculate the average value. Then subtract the average value of the satellite from the primary difference of the satellites, make the secondary difference to obtain the error of the augmented satellite offset. Finally, the RMS value of the satellite clock offset error is calculated.

Accuracy analysis of SIS User Range Error (SISURE) : after the accuracy analysis of the orbit and clock offset, the following formula was used to calculate the SISURE of different satellites, and finally the RMS value of SISURE of each satellite was calculated.

$$SISURE = \sqrt{(\omega_R \Delta R - cdt)^2 + \omega_{A,C} (\Delta A^2 + \Delta C^2)} \quad (2.4)$$

In Equation (2.4), ω_R and $\omega_{A,C}$ are the contribution factors of radial direction R and along-track direction A and cross-track direction C, c are the speed of light, dt are the error of satellite clock, and $\Delta R, \Delta A, \Delta C$ are the errors in the radial direction, along-track direction and cross-track direction of orbit, respectively.

The final orbit and clock offset of QZSS were selected as a reference. The results of comparing the satellite clock after L6 augmentation with the QZSS final clock are shown in Fig. 2 and Fig. 3, and the errors of the orbit, clock and SISURE are shown in Fig. 4.

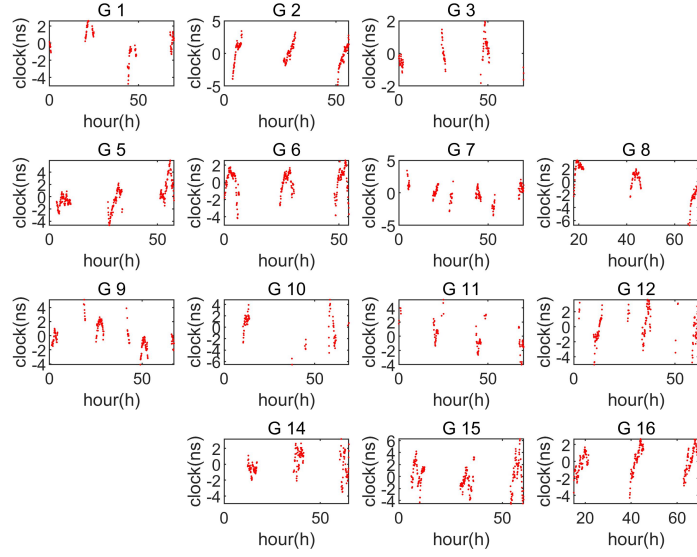


Fig. 2. GPS 1-16 star clock difference comparison

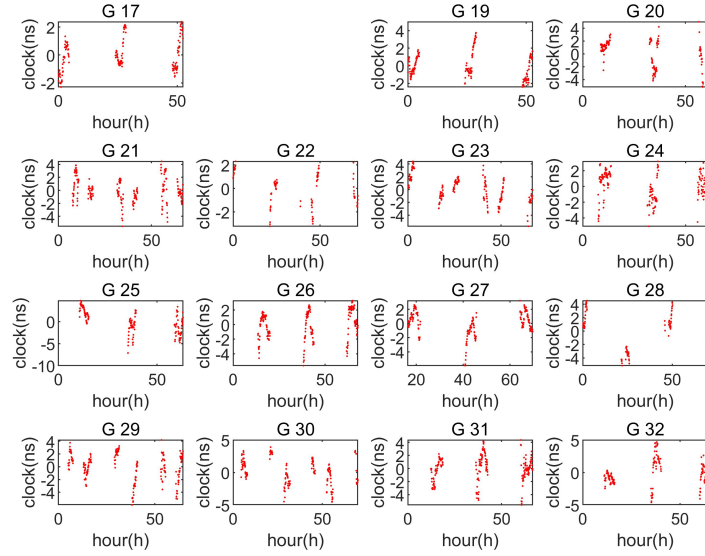


Fig. 3. GPS 17-32 star clock difference comparison

In the Fig. 2 and Fig. 3, G04, G13 and G18 satellites are missing because the orbit information of these three satellites is not published in the precise ephemeris

of QZSS. The clock difference comparison sequence diagram is discontinuous because L6 products only release satellite correction information that is observable and available in Japan. It can be seen from Fig. 4, the clock error RMS is between 0.15m and 0.65m. The orbit error fluctuates greatly, and the radial direction is generally below 0.65m. The G10 satellite has a large error, which exceeds 1m. Along-track direction and cross-track direction fluctuate greatly, leading to a maximum of 2.2m. SISURE as a whole was within 0.9m, while G10 and G21 satellites reached 1.7m and 1.2m, respectively. The G10 and G21 satellites fluctuated significantly compared to other satellites, and the correction errors were relatively large. The corrections of other satellites were relatively stable. On average, the augmented clock offset accuracy is about 0.46m, radial average accuracy is about 0.26m, along-track accuracy is about 1.05m, cross-track accuracy is about 0.94m, three-dimensional accuracy is about 1.44m, and SISURE accuracy is about 0.62m. In general, there is still a certain gap between the augmented GPS orbit and clock offset and the QZSS final orbit and clock offset. Analyzing the reasons, the CLAS service of QZSS is oriented to PPP-RTK service, and the clock offset and orbit correction information published at each moment are not precise orbit and clock offset correction information. Compared with the correction information issued by the IGS Real-Time Service (IGS-RTS) at the same time, there are certain errors in the corrections and even the opposite corrections, but this in CLAS service is normal, because the CLAS augmented service emphasizes the sum of the corrections in the direction of the receiver-to-satellite line of sight (including orbit correction, clock difference correction, satellite code and phase bias correction, etc.) is always the best. Therefore, the accuracy of the orbit and clock offset may not be ideal after correction.

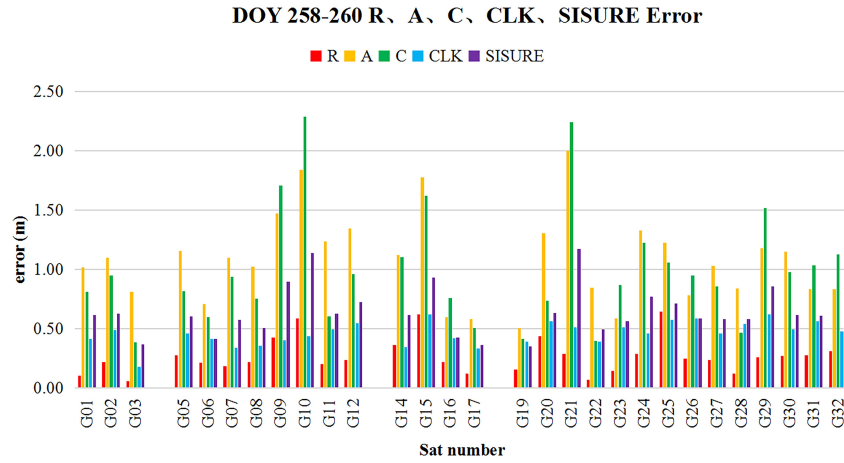


Fig. 4 . GPS 1-32 star R, A, C, CLK, SISURE error

4 Precise Point Positioning analysis of L6D products based on QZSS system

In order to evaluate the PPP augmentation performance of L6D augmentation information, this paper selected the MGEX (Multi-GNSS Experiment) observation data of the four stations in Japan: AIRA, STK2, TSK2, and USUD. L6 products published by QZSS were used to perform a post static PPP processing of DOY 258-260 on the four stations. At the same time, the precise orbit and precise clock products published by QZSS were used for the same processing of the above data, and the processing results were taken as reference. Finally, the coordinates in the SINEX file of IGS on that day were used as the real value to calculate deviation statistics of PPP results in the directions of E, N and U.

Table 2. PPP treatment plan

Project category	CLAS plan	QZSS plan
Orbit product	Brdc+CLAS correction	QZSS final product
Clock product	Brdc+CLAS correction	QZSS final product
Ionospheric error	Dual-frequency Iono-free	Dual-frequency Iono-free
Tropospheric error	CLAS correction	Saastamoinen model
Code bias	CLAS correction	Code DCB correction
Phase bias	CLAS correction	null
Cut-off height Angle/(°)	15	15
Tidal correction	Model correction	Model correction
Satellite antenna phase center correction	Model correction	Model correction
Receiver antenna phase center correction	Model correction	Model correction
Receiver position	SPP calculation	SPP calculation
Ambiguity	Not fixed	Not fixed

Table 3. Introduction of using station information

station name	Network ID	Receiver type	Receiver antenna type	Received data type
AIRA	3	Trimble NetR9	TRM59800.00	G+E+R+J
STK2	10	Trimble NetR9	TRM59800.00	G+E+R+J
TSK2	7	Trimble NetR9	TRM59800.00	G+E+R+J
USUD	7	SEPT POLARX5	AOAD/M_T	G+E+R+C+J

Table 2 shows the two PPP treatment plans, and Table 3 shows the relevant information of test stations used in the experiment. The post static PPP processing

was performed on the four stations for three days. Fig. 5 shows the deviation of the positioning results of the two plans of the four stations in the directions of E, N and U.

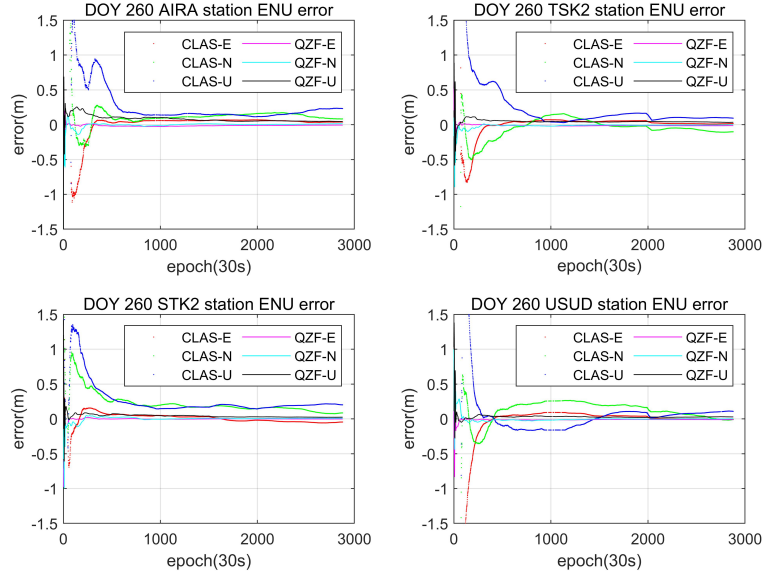


Fig. 5. Positioning accuracy of four stations in DOY 260

Table 4 shows the statistical results of PPP positioning accuracy after CLAS augmentation and PPP positioning accuracy using QZSS final precise ephemeris and clock products. It can be seen that the positioning error after the augmentation is smaller in the direction of E, at the level of cm; However, it fluctuates greatly in the direction of N and in the level of centimeter-decimeter. The horizontal accuracy of the four stations fluctuates between centimeters and decimeters. In Fig. 5, it can be clearly found that the convergence rate of positioning using QZSS final precise ephemeris and clock product is significantly faster than that of CLAS augmented positioning. As can be seen from Fig. 5, the positioning result after CLAS augmentation is about 6cm in the direction of E, 11cm in the direction of N, and 18cm in the direction of U. The positioning results of the final precise ephemeris and clock products using QZSS are better than the augmented results. The DOY 258-260 positioning process was performed for three days, and values with deviations less than 20cm in the direction of E and N and less than 40cm in the direction of U were selected for statistics. The positioning direction of ENU in CLAS plan converges to 20cm, 20cm and 40cm at about 420 (30s) of the period, and the RMS for calculating the deviation values of the three directions are shown in Table 4. The PPP positioning accuracy after CLAS augmentation can achieve centimeter-decimeter level positioning, horizontal accuracy is about 13cm, vertical

accuracy is about 16cm, and L6 file can be used for augmented positioning service. However, the results of PPP positioning with QZSS final precise ephemeris and clock product still have large errors. This is because CLAS is oriented to the PPP-RTK service, and there is a difference between the algorithms of PPP and PPP. Therefore, when performing PPP positioning, there may be large errors in the positioning results. When CLAS is applied to the PPP-RTK service, it can achieve higher positioning accuracy and achieve centimeter-level positioning accuracy.

Table 4. RMS statistics of PPP positioning errors for two plans of four stations

Station	CLAS- E(cm)	CLAS- N(cm)	CLAS- U(cm)	CLAS- H(cm)	QZF- E(cm)	QZF- N(cm)	QZF- U(cm)	QZF- H(cm)
AIRA	7.85	8.17	16.87	11.33	0.74	1.11	4.99	1.33
TSK2	5.13	12.44	17.94	13.46	0.41	0.90	1.70	0.99
STK2	3.48	8.89	12.22	9.54	0.58	0.82	3.60	1.00
USUD	7.92	14.96	17.57	16.93	0.48	0.42	1.78	0.64
mean	6.09	11.12	16.15	12.68	0.55	0.81	3.02	0.99

5 Conclusion

In this paper, the performance of QZSS L6D augmentation information is evaluated, and uses the QZSS final orbit and clock products as references to evaluate the augmentation effect of L6D signals. The results show that the L6D augmentation information of QZSS can provide the augmentation service. After the augmentation, the clock offset precision is about 0.46m, the radial average precision is about 0.26m, the along-track precision is about 1.05m, the cross-track precision is about 0.94m, the 3D precision is about 1.44m, and the SISURE precision is about 0.62m. The static PPP positioning horizontal accuracy based on QZSS L6D augmentation information is about 13cm and the vertical accuracy is about 16cm, which still has a large error with the final precise ephemeris and precise clock PPP positioning results of QZSS. Further research is needed on the PPP-RTK performance of QZSS L6D augmentation information and the PPP performance of L6E augmentation information.

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